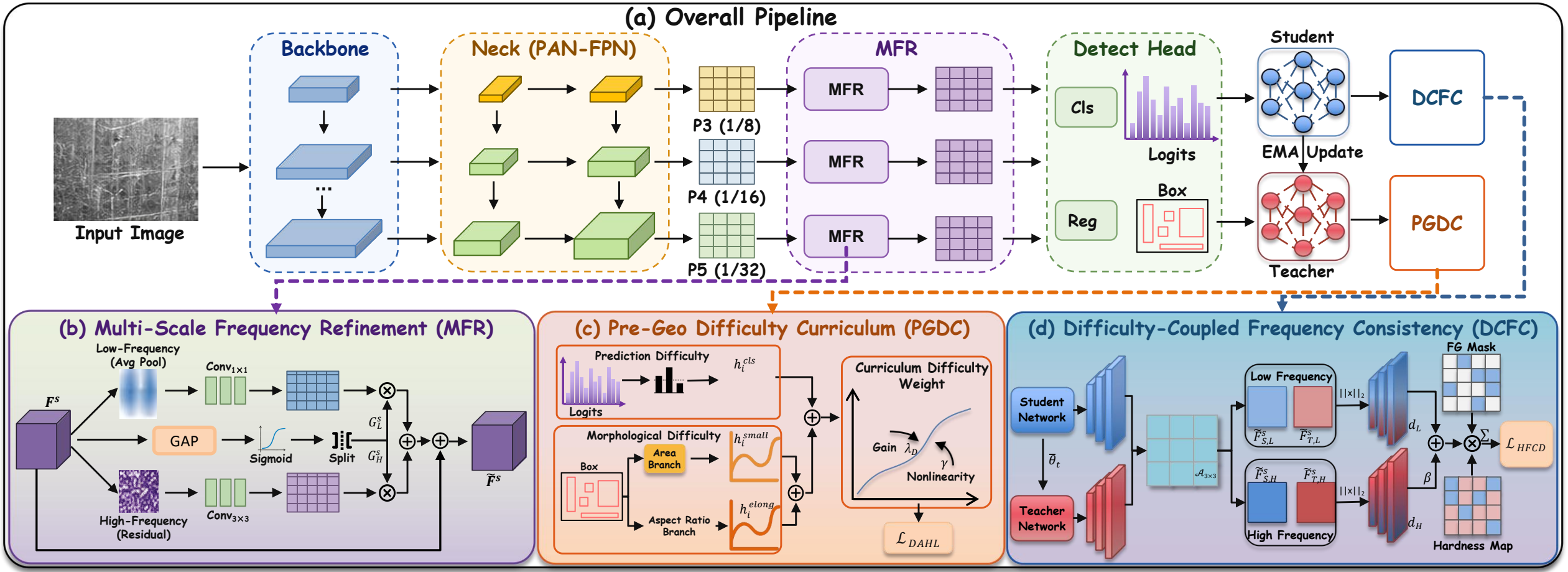

(a) Overall Pipeline
Input Image
Backbone
Neck (PAN-FPN)
P3 (1/8)
P4 (1/16)
P5 (1/32)
MFR
MFR
MFR
MFR
Detect Head
Cls
Logits
Reg
Box
Student
EMA Update
Teacher
DCFC
PGDC
(b) Multi-Scale Frequency Refinement (MFR)
Low-Frequency (Avg Pool)
Conv1×1
GAP
Sigmoid
Split
High-Frequency (Residual)
Conv3×3
(c) Pre-Geo Difficulty Curriculum (PGDC)
Prediction Difficulty
Logits
Morphological Difficulty
Box
Area Branch
Aspect Ratio Branch
Curriculum Difficulty Weight
Gain
Nonlinearity
(d) Difficulty-Coupled Frequency Consistency (DCFC)
Student Network
Teacher Network
Low Frequency
High Frequency
FG Mask
Hardness Map